

Table 1: Key libraries for data engineering, analytics and visualisation

Library	Programming language	Implementation category	Key application segment
Lux	Python	Auto visualisation	Smart insights
Pandas	Python	Data wrangle	Tabular analysis
NumPy	Python	Numeric compute	Fast arrays
SciPy	Python	Scientific tools	Stats suite
Matplotlib	Python	Data plotting	Base charts
Seaborn	Python	Statistical plots	Pretty plots
Plotly	Python/JS	Interactive charts	Web visuals
Bokeh	Python	Interactive plots	Browser output
Altair	Python	Declarative visualisation	Grammar charts
scikit-learn	Python	Machine learning	ML models
TensorFlow	Python	Deep learning	Neural nets
PyTorch	Python	Deep learning	Research DL
Keras	Python	High-level DL	Easy neural
XGBoost	Python/R	Gradient boost	Boost trees
LightGBM	Python/R	Gradient boost	Fast trees
CatBoost	Python/R	Gradient boost	Categorical boost
Statsmodels	Python	Econometrics	Statistical tests
Dask	Python	Parallel compute	Scale Pandas
Vaex	Python	Large DataFrames	Out-of-core
Polars	Python/Rust	Fast DataFrames	Column engine
DuckDB	Multi-language	In-process SQL	OLAP query
Arrow	Multi-language	Column format	Memory spec
SQLite	C/Python	Embedded SQL	File DB
Apache Spark	Scala/Python	Big Data	Cluster compute
Apache Flink	Java/Scala	Stream engine	Realtime data
Hadoop MapReduce	Java	Batch engine	Distributed jobs
Hive	Java	SQL engine	Big SQL
Pig	Java	Data flows	Scripting ETL
Julia DataFrames	Julia	DataFrames	Tabular ops
Julia Stats	Julia	Statistics	Stats tools
Plots.jl	Julia	Data plotting	Multi-backend
R dplyr	R	Data wrangle	Tidy tools
R ggplot2	R	Data plotting	Grammar graphics
R data.table	R	Fast tables	High-speed ops
R caret	R	Machine learning	ML suite
R Shiny	R	Web apps	Interactive dash
Weka	Java	Machine learning	Classic ML
Deeplearning4j	Java	Deep learning	JVM neural
RAPIDS cuDF	Python	GPU DataFrame	GPU Pandas
cuML	Python	GPU ML	GPU models
Orange	Python	Visual ML	GUI ML
KNIME	Java	Visual analytics	Drag workflow
Octave	Octave	Numeric compute	MATLAB-like
Scilab	Scilab	Numerical suite	Scientific tools
Gretl	C/GUI	Econometrics	Stats models
Giotto-tda	Python	Topological ML	TDA tools
MLJ.jl	Julia	Machine learning	Unified ML
Rust Polars	Rust	DataFrames	Fast column
Vega-Lite	JS	Declarative visualisation	JSON spec
D3.js	JS	Data visuals	SVG charts
Chart.js	JS	Web charts	Simple charts
Apache Superset	Python	BI dashboard	Web BI